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CONDUCTED BY R. CHAMBERS (SECUNDUS)

No. 152.—VOL. III.

SATURDAY, NOVEMBER 27, 1886.

PRICE 1½d.

SEALSKIN COATS, ALIVE AND DEAD.

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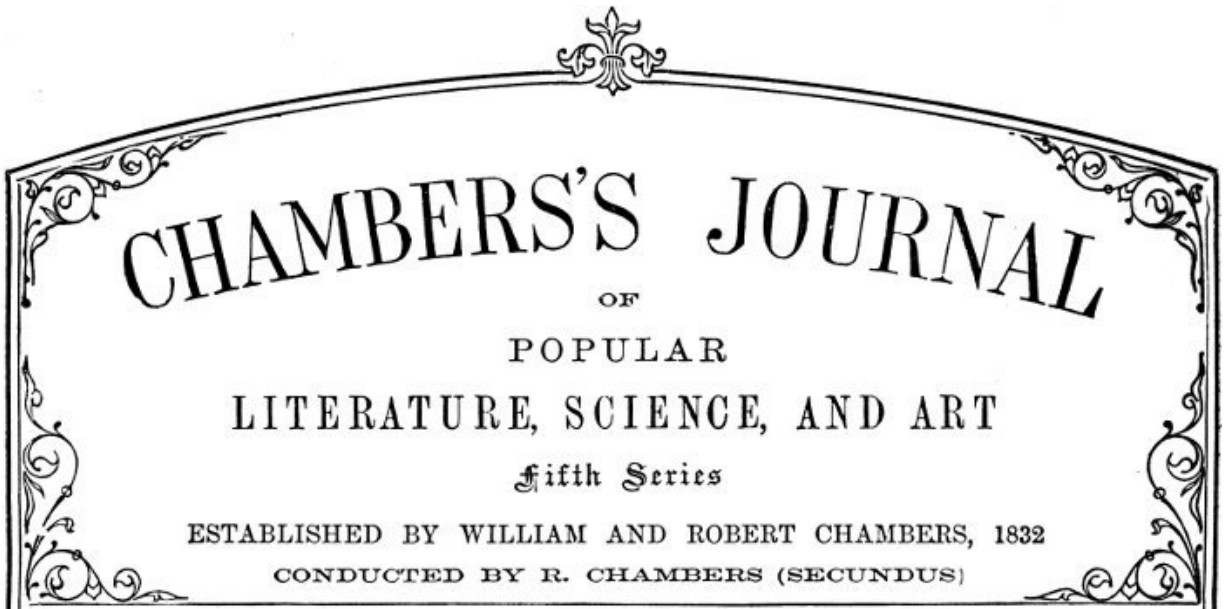
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The female fur-seal is from four to four and a half feet in length from snout to tail, lithe in form, without the heavy covering of fat round the shoulders which the male has, and with beautiful, gentle, intelligent, dark-blue eyes.

She will weigh from fifty to a hundred pounds, according to her condition. Her manners are as amiable as her eyes, and she never fights with her neighbours, as her quarrelsome lord and master does. The cow-seal has but one voice—a sort of bleating half-way between the cry of a calf and that of an old sheep—and this is used for calling the young, which, curiously enough, are known as ‘pups,’ although the mothers are ‘cows,’ and the fathers ‘bulls.’ The male seal, however, has four voices. One is for battle, and resembles the puffing of a labouring locomotive; another is a hoarse loud roar; a third is a sort of low gurgle or growl; and a fourth, a sort of chuckle, half-hiss, half-whistle. The breeding-grounds are called ‘rookeries,’ and there, during the season, the din of roars, puffs, growls, and whistles from countless thousands of vigorous ‘bulls,’ is ceaseless, and in volume has been compared to the boom of Niagara.

It is odd that the breeding-place of ‘bulls’ and ‘cows’ should be called ‘rookeries,’ but so it is. The first to arrive at these rookeries are the bull-seals, and the season begins about the first of May. As it is ‘First come, first served,’ and as there is an unwritten law among them that a bull requires a clear space of from six to eight feet square for the accommodation of himself and family, there is much scrambling and fighting for plots, and the late arrivals may be driven away without being allowed a landing-place at all. They fight with great strength and courage—only the adult males, however—running at each other with averted heads, and then seizing each other with their teeth. The battles are often long, and the wounds severe; but these soon heal; and an adventurous ‘bull’ thinks nothing of forty or fifty desperate combats in a season. While fighting, they utter both their roar and their whistle, the hair is sent flying in all directions, and the eyes gleam with angry fire. It is said that in a seal-fight there is always an offensive and a defensive party, and that if the latter is beaten, he simply vacates his position to the victor, who does not follow his foe, but lies down on the conquered territory and gives vent to his chuckle.

Although the cows are amiable, they are not particularly demonstrative to their infants, which are born immediately after the females are located in the rookeries. Twins are very rare, and mothers always suckle their own young. The pups do not know their own mothers, and if separated from them, will take with the greatest alacrity to the first kindly cow which will console them with her rich creamy and abundant milk. The pups, for the

first three months after birth, are jet black in colour, and bleat in a minor key after the fashion of the cows. At birth, a pup will weigh three or four pounds, and measure twelve or fourteen inches in length. Curiously enough, the pup-seal cannot swim, and even if he is several weeks old, will helplessly sink, if thrown into the water. But about the second week of August begins one of the most curious episodes of seal-life—the education of the young. By the time he has counted six weeks or so of life, the pup-seal begins to feel an inclination to play on the margin of the sea, where, as the waves flow and recede, the shore is alternately covered and uncovered. The baby-seal finds that thousands and thousands and tens of thousands of his fellow-babies have been smitten with the same curiosity about the sea almost simultaneously with himself, and that the beach is swarming with tumbling, floundering, gurgling, whistling, playful, yet nervous young animals. By-and-by, one plucks up courage to try a plunge in the deeper surf; others follow; one gets carried beyond his depth, and in frantic struggles to reach the shore again, discovers that he has a power of locomotion even in the water. It is but feeble; and when a kindly wave chucks him out of harm's way on to the rocks, he is blown and exhausted. But he takes a short sleep, and then has another go; and after a few more efforts, finds, to his great delight, that he is even more at home in the water than on the land. For the next few weeks the coast-waters of the islands are black with the little fat bodies revelling in their new-found power, and gamboling among the breakers like children on the grass. It used to be believed by the old sailors that the parent seals drove their young ones into the water and taught them forcibly to swim; but more recent and careful observation places it beyond doubt that the parents take no part whatever in the process of education, but leave the young ones to learn the battle of life for themselves.

By the time the breeding season is over, all the young seals have become able-bodied swimmers. By this time, too, the pups have grown to thirty or forty pounds-weight, and have changed the black coat of infancy for the thick, gray, hairy coat of youth. At this age, the coats of both male and female are similar; indeed, not until the third year do they assume their permanent differences. The outer coat of the full-grown bull is of a dark-brown colour, and the hairs are short and crisp; beneath, like the down under the feathers of a bird, is the close, soft, elastic fur, so esteemed by man, or rather woman. The full-grown cows, as they come into the

rookeries at the beginning of the season, are of a dull, dirty-gray colour, which, after they have been a short time on land, changes to a rich steely gray on the back, and snow-white on the chest and belly; but after a few weeks the white changes into a dull ruddy colour, and the steel gray into a brownish gray. The breeding season is over by the end of July; the families begin to break up, and the rookeries to be disorganised during August. By the middle of September, all order and distinction is lost, and the young ones have commenced life on their own account. By the end of October, all the mature seals have left the islands; and by the end of November, even the youngest have disappeared.

Whither? That is one of the conundrums of nature, as is also the question, where do the seals die? It is certain that none perish from natural causes on the islands, and all that is known of their doings elsewhere is, that they seem usually to shape a southern course. They are lost in the vast mazes of the Pacific, not to be seen of man again until the following summer. They have natural enemies in sharks and other submarine animals of prey; but it is not thought that their numbers suffer much diminution on this account. Their own food is fish, and Mr Elliott has calculated that an adult male seal will consume forty pounds, and an adult female ten to twelve pounds, per day, of fresh fish. Taking, with the young ones, an average of ten pounds per day each, and the numbers annually frequenting the rookeries of the Pribylov Islands—which have been ascertained by careful measurement and estimate at about four millions and three-quarters—we have a total of six millions of tons of fish consumed every year by the fur-seals! The figures are stupendous, but they seem beyond doubt.

As to the now approximately known number of seals, there is no reason to believe that it is any greater than it was when the islands were first discovered; and while the number will not be decreased by the present method of capture, it is not thought that it will increase. The supply of fur-seals, then, may be taken as a fixed quantity, with a known annual yield to man. That yield is restricted by the law of the United States to one hundred thousand skins per annum. The government holds the islands for the State and leases the right of capture to a Company, who are permitted not to take a larger number than that just mentioned. They employ the natives of the Aleutian Islands, who work in gangs, under their chiefs, and receive forty cents, or one shilling and eightpence, for every 'pelt' or hide they hand to

the Company's officials. Government officers, again, keep a separate tally; so there is a double check upon the Company, who cannot easily, even if they wish, exceed their prescribed rights. As the annual birth-rate is about one million, of which one half are males, the number annually abstracted by man can have no appreciable effect in reducing the supply or in affecting the natural increase. The average natural life of the male seal is believed to be from fifteen to twenty years, and that of the female, about ten years, so that deaths by man on the rookeries, and from submarine foes during the winter, suffice to keep the race within the bounds now known.

The men operate only on the haunts of the 'bachelor' seals. It is presumed that about two-thirds of the males are not allowed to land on the rookeries by the stronger and abler remanent, so that the wants of man can be supplied without interfering with the operations of the breeding-grounds. When the 'bachelors' are dozing about the shores in the early summer, the natives get in quietly between them and the sea. The seals on perceiving the men turn to run inland, and are easily driven to the appointed killing-grounds. Three or four men can easily guide and secure as many thousand seals, and the driving is done leisurely, for if the animals become overheated, the fur is injured. The men therefore allow them to rest from time to time, and renew the drive by clattering and shouting, to startle the seals to fresh exertions. They move with the docility of a flock of sheep, and only the old bulls ever show fight. These last will occasionally make a stand and act on the defensive; but as they are of little value commercially, the bellicose oldsters are allowed to drop out and go their own ways. It is only the animals between one and five years old which are desired, for after the fifth year, the fur deteriorates, the undergrowth becoming shorter and coarser. The thickest and finest pelts are those of the third and fourth years. Beneath the skin is a dense layer of oily blubber, which, unlike the blubber of the hair-seal, has a very offensive odour.

The work of catching and pickling the pelts occupies June and July, by which time the Company will have secured its legal number of one hundred thousand, or as many short of the number as circumstances have confined them to. After July, the seals begin to moult, and the skins become of less and less value as the season advances. Altogether, three hundred and ninety-eight persons are employed annually on the Pribylov Islands in this work.

After the 'catch' is ended, the skins are taken in the Company's steamers to San Francisco, and thence nearly all or about nine-tenths are shipped to London, for London has the monopoly of the preparation of these furs for market. The skins as they come into England are very different in appearance from what we see on the backs of our lady-friends. They are indeed very unattractive; and all the coarse stiff outer hair has to be carefully extracted before the rich under-fur is seen. This last is then dyed and dressed. It is hurried or defective dyeing and dressing which accounts for the variation in prices of the finished furs, for there is little difference in the original quality. The more careful and skilful the work of the furrier, therefore, the dearer becomes the sealskin jacket.

The Alaska Commercial Company's lease of the islands is for twenty years from the 1st of May 1870, and they pay the government a rental of eleven thousand pounds per annum for the islands, and a tax of eight shillings for each sealskin, ten and sixpence for each fur-seal skin, and fifty-five cents for every gallon of oil, shipped. The Company is also bound to supply the inhabitants with a stipulated quantity of dried fish, firewood, and salt; to maintain a school on each island for the education of the natives; and not to sell or give any 'distilled spirituous liquors' to the natives. We believe that the Company has in only one year (1881) taken its full number of skins, the usual number shipped being from ninety to ninety-five thousand. Between 1870 and 1881, the Company had paid the United States Treasury nearly three and a half millions of dollars in rent and royalty.

BY ORDER OF THE LEAGUE.

CHAPTER XVI.

CONSUMED by conflicting emotions, and torn by a thousand hopes and fears, Maxwell set out on his journey to Rome. At any hazards, he was determined to commit no crime, and trusted to time and his own native wit to show him a way out of the awful difficulty which lay before him. All the old familiar country he passed through failed to interest him now; he saw nothing but his own fate before his eyes; and the Eternal City, which had once been a place of mystery and delight to him, now looked to his distorted fancy like a tomb, every broken statue an avenging finger, and every fractured column a solemn warning.

It was night when he arrived and secured apartments—the old ones he had occupied in his student days, the happiest time in his life, he thought now, as every ornament recalled this silent voice or that forgotten memory slumbering in some corner of his brain. He could eat nothing; the very air of the place was oppressive to him; so he put on his hat and walked out into the streets, all alive with the citizens taking their evening walk, and gay with light laughter over flirtations and cigarette smoke. He wandered long and far, so far, that it was late when he returned; and there, lying on the table, was a sealed packet, bearing the device of the Order, and in the corner two crossed daggers. He groaned as he opened it, knowing full well the packet contained the hated ‘instructions,’ as they were called. He tore them open, read them hastily, and then looked out of the window up to the silent stars. And it was Visci, his old friend Carlo Visci, he was sent here—to murder! The whole thing seemed like a ghastly dream. Visci, the truest-hearted friend man ever had; Visci, the handsome genius, whose purse was ever ready for a fellow-creature in need; the man who had sat at his table times out of number; the student who was in his secrets; the man who had saved his life, snatched him from the very jaws of death—from the yellow waters of the Tiber. And this was the friend he was going to stab in the back some dark night! A party of noisy, light-hearted students passed down the street, some English voices amongst them, coming vaguely to Maxwell’s

ears, as he sat there looking on the fatal documents, staring him in the face from the table.

‘Et tu, Brute!’

Maxwell looked up swiftly. And there, with one trembling forefinger pointing to the open documents, stood the figure of a man with a look of infinite sorrow on his face, as he gazed mournfully down upon the table. He was young—not more than thirty, perhaps, and his aquiline features bore the marks of much physical suffering. There were something like tears in his eyes now.

‘Carlo! is it possible it is you?’ Maxwell cried, springing to his feet.

‘Yes, Fred, it is I, Carlo Visci, who stand before you. We are well met, old friend; you have not far to seek to do your bidding now. Strike! while I look the other way, for it is your task, I know.’

‘As there is a heaven above us, no!’ Maxwell faltered. ‘Never, my friend! Do you think I would have come for this? Listen to me, Visci. You evidently know why I am here; but sure as I am a man, never shall my hand be the one to do you hurt. I have sworn it!’

‘I had expected something like this,’ Visci replied mournfully. ‘Yes, I know why you came. You had best comply with my request. It would be a kindness to me to kill me, as I stand here now.’

‘Visci, I swear to you that when I joined the Brotherhood, I was in the blackest ignorance of its secret workings. When I was chosen for this mission, I did not even comprehend what I had to do. Then they told me Visci was a traitor. Even then, I did not know it was you. Standing there in the room, I swore never to harm a hair of your head; and, heaven help me, I never will!’

‘Yes, I am a traitor, like you,’ Visci smiled mournfully. ‘Like you, I was deceived by claptrap talk of liberty and freedom; like you, I was allotted to take vengeance on a traitor; and like you, I refused. Better the secret dagger than the crime of fratricide upon one’s soul!’

‘Fratricide! I do not understand.’

‘I do not understand either. Frederick, the man I was detailed to murder—for it is nothing else—is my only brother.—You start! But the League does

not countenance relationships. Flesh and blood and such paltry ties are nothing to the friends of liberty, who are at heart the sternest tyrants that ever the mouth of man execrated.—But what brings you here? You can have only one object in coming here. I have told you before it would be a kindness to end my existence.’

‘But why? And yet, when I come to look at you again, you have changed.’

‘I have changed,’ Visci echoed mournfully—‘changed in mind and body. My heart is affected, diseased beyond all hope of remedy. I may die now, at any moment; I cannot live four months.’

They sat down together, and fell to discussing old times when they were happy careless students together, and Maxwell did not fail to notice the painful breathing and quick gasping spasms of his friend, altered almost beyond recognition from the gallant Visci of other days.

‘Salvarini advised me to come here. You remember him; he claims to be a true friend of yours,’ Maxwell observed at length. ‘He said it would gain time, and enable me to form my plans.—But tell me how you knew I was in Rome. I have only just arrived.’

‘I had a sure warning. It came from the hand of Isodore herself.’

‘I have heard much of her; she seems all-powerful. But I thought she was too stern a Leaguer to give you such friendly counsel. Have you ever seen her? I hear she is very beautiful.’

‘Beautiful as the stars, I am told, and a noble-hearted woman too. She is a sort of Queen of the League; but she uses her power well, ever erring on the side of mercy. She has a history, report says—the old story of a woman’s trustfulness and a man’s deceit. Poor Isodore! hers is no bed of roses!’

‘And she put you on your guard?’ Maxwell asked. ‘Come, there must be some good in a woman like that, though I cannot say I altogether like your picture. I should like to see her.’

‘I should not be surprised if you did before many days. She is the one to protect you from violence. With her sanction, you could laugh the mandates of the League to scorn. Had I long to live, I should sue for her protection, and wherever she may be, she would come to me. Even now, if she comes to Rome, see her if you can and lay your case before her.’

‘And shield myself behind a woman! That does not sound like the chivalrous Visci of old. She is only a woman, after all.’

‘One in a million,’ Visci answered calmly. ‘If she holds out her right hand to you, cling to it as a drowning desperate man does to a rock; it is your only chance of salvation.—And now it is late. I must go.’

Despite his own better sense, Maxwell began to dwell upon the fact of gaining assistance from the mysterious Isodore. At meetings of the League in London, he had heard her name mentioned, and always with the utmost reverence and affection. If she could not absolutely relieve him from his undertaking, she could at anyrate shield him from non-compliance with the mandate. Full of these cheerful thoughts, he fell asleep.

He found his friend the following morning quite cheerful, but in the daylight the ravages of disease were painfully apparent. The dark rings under the eyes and the thin features bespoke nights of racking pain and broken rest.

Visci noticed this and smiled gently. ‘Yes, I am changed,’ he said. ‘Sometimes, after a bad night, I hardly know myself. It is cruel, weary work lying awake hour after hour fighting with the grim King. But I have been singularly free from pain lately, and I am looking much better than I have been.’

‘There might be a chance yet,’ Maxwell replied with a cheerfulness wholly assumed, and thinking that this ‘looking better’ was the nearest approach to death he had ever seen. ‘An absence from Rome, a change of climate, has done wonders for people before now.’

Visci shook his head. ‘Not when the mainspring of life is broken,’ he said: ‘no human ingenuity, no miracle of surgery can mend that. Maxwell, if they had deferred their vengeance long, they would have been too late. Some inward monitor tells me I shall fail them yet.’

‘You will for me, Visci, you may depend upon that. Time is no object to me.’

‘And if I should die and disappoint you of your revenge, how mad you would be!’ Visci laughed. ‘It is a dreadful tragedy to me; it is a very serious thing for you; and yet there is a comic side to it, as there is in all things. Ah me! I cannot see the droll side of life as I used; but when the bloodthirsty

murderer sits down with his victim tête-à-tête, discussing the crime, there is something laughable in it after all.'

'I daresay there is,' Maxwell answered grimly, 'though I am dense enough not to notice it. To me, there is something horribly, repulsively tragic about it, even to hear you discussing death in that light way.'

'Familiarity breeds contempt. Is not that one of your English proverbs?' Visci said airily.—'But, my good Frederick,' he continued, lowering his voice to a solemn key, 'the white horseman will not find me unprepared, when he steals upon me, as he might at any moment. I am ready. I do not make a parade of my religion, but I have tried to do what is right and honest and honourable. I have faced death so often, that I treat him lightly at times. But never fear that when he comes to me for the last time'——

Maxwell pressed his friend's hand in silent sympathy. 'You always were a good fellow, Visci,' he said; 'and if this hour must come so speedily, tell me is there anything I can do for you when—when'——

'I am dead? No reason to hesitate over the word. No, Maxwell; my house is in order. I have no friends besides my brother; and he, I hope, is far beyond the vengeance of the League now.'

'Then there is nothing I can do for you in any way?'

'No, I think not. But you are my principal care now; your life is far more important than mine. I have written to Isodore, laying a statement of all the facts before her; and if she is the woman I take her for, she is sure to lose no time in getting here. Once under her protection, you are safe; there will be no further cause for alarm.'

'But it seems rather unmanly,' Maxwell urged.

'Unmanly!' echoed Visci scornfully. 'What has manliness to do with fighting cowardly *vendetti* in the dark? You must, you shall do it!' he continued vehemently; but the exertion was too much for him, and he swayed forward over the table as if he would fall. Presently, a little colour crept into the pallid face, and he continued: 'You see, even that is too much for me. Maxwell, if you contradict me and get me angry, my blood will be upon your head after all. Now, do listen to reason.'

‘If my want of common-sense hurts you as much as that, certainly. But I do not see how this mysterious princess can help me.’

‘Listen to me,’ Visci said solemnly. Then he laid all his schemes before the other—his elaborate plans for his friend’s safety, designs whose pure sacrifice of self were absolutely touching.

Maxwell began to take heart again. ‘You are very good,’ he said gratefully, ‘to take all this infinite pains for me.’

‘In a like strait you would do the same for me, Fred.’

‘Yes,’ Maxwell answered simply. ‘How Salvarini’s words come back to me now! Do you remember, when I wanted to throw my insignia out of the window that evening, the last we all spent together?’

‘I recollect. It was two days before little Genevieve disappeared,’ Visci answered sadly.—‘Do you know, I have never discovered any trace of her or Lucrece. Poor child, poor little girl! I wonder where she is now.’

‘Perhaps you may see her again some day.’

‘It has long been my dearest wish; but it will never be fulfilled now. If ever you do see her once more, say that I’——

‘Visci!’

As the last words fell from the Italian’s lips, his head hung forward, and he fell from his chair. For a moment he lay motionless, then raised his face slightly and smiled. A thin stream of blood trickled down his fair beard, staining it scarlet. He lay quietly on Maxwell’s shoulder.

‘Do not be alarmed,’ he said faintly. ‘It has come at last.—There are tears in your eyes, Fred. Do not weep for me. Do not forget Carlo Visci, when you see old friends; and when you meet little Genevieve, tell her I forgave her, and to the last loved and grieved for her.—Good-bye, old friend. Take hold of my hand. Let me look in your honest face once more. It is not hard to die, Fred. Tell them that my last words——Jesu, mercy!’

‘Speak to me, Carlo—speak to me!’

Never again on this side of the grave. And so the noble-hearted Italian died; and on the third day they buried him in a simple grave under the murmuring pines.

No call to remain longer now. One last solitary evening ramble, Maxwell took outside the city wall ere his departure. As he walked along wrapped in his own sad thoughts, he did not heed that his footsteps were being dogged. Then with a sudden instinct of danger, he turned round. The feet that followed stopped. 'Who is there?' he cried.

A muffled figure came towards him, and another stealthily from behind. A crash, a blow, a fierce struggle for a moment, a man's cry for help borne idly on the breeze, a mist rising before the eyes, a thousand stars dancing and tumbling, then deep, sleepy unconsciousness.

(To be concluded next month.)

THE PLEASURES OF RUIN.

THERE must be many people to whom the above heading will be at once suggestive of the famous chapter upon Snakes in Iceland; but to the philosophical mind—and it is marvellous how philosophical one can become under adversity—there are certain compensating advantages in the state of ruin, which, if not quite so intense as the Pleasures of Hope, or Memory, or Imagination, do much to reconcile us to the change in our circumstances. The first feeling is one of extreme relief that the whole thing is over and we are out of suspense. The smash has come; writs and summonses have blossomed into sheriffs' officers, and the auctioneer, whose fell and inexorable hammer has made short work of our goods and chattels; our wealthy friends have said that they knew it would come to this; and Jones, who used to look dinners and five-pound notes at us whenever he met us formerly, now crosses over to the opposite side of the street. The cheap lodgings in the shady neighbourhood have become hard and ineradicable facts, and we can look about us at last and endeavour to make the best we can of the position.

You now have a newly acquired sense of freedom and independence to which perhaps you have long been a stranger. It is no longer a question of whether you shall dine at the *Bristol* or the *Blue Posts*, but in all likelihood the choice will lie between the *diner du jour* in Leicester Square, a chop, or Duke Humphrey. Nor, if you be a married man, need you now vex your soul with the proper precedence of a brigadier-general, an Indian judge, a colonial bishop, and a resident commissioner from the Punjab, as has happened in the days gone by when you gave a dinner. Nor will the varying merits of asparagus soup and turtle, salmon mayonnaise and aspic of lobster, truffled turkey and oyster-stuffed capon, and all the rest of it, come between you and your night's rest. Again, your circumstances are such that you are no longer harassed by the touters for subscriptions, male and female, and you find it therefore needless to discuss the comparative merits of the claims put forward by the friends of the Cannibal Islanders for French mustard, and by the friends of the Mayor of Little Pedlington for a

new pump in the market-place in honour of that excellent cheesemonger and municipal chief.

When you go to the theatre or opera, you are no longer compelled to pay fifty or a hundred per cent. for the privilege of receiving your ticket from an agent, and you go to the pit, where, if the orange peel and ginger beer and nuts are a bit of a nuisance at first, you are not long in getting used to it; and at anyrate you are permitted to hear the piece without being bored by one of Smith's 'good stories' during Patti's chief *aria*, or while Irving is giving some fine piece of declamation. You discover sources of gratuitous amusement which indifference has hitherto hidden from you. That glorious rotunda in Bloomsbury, the British Museum Reading-room—the mausoleum of the mind of the world—gives you opportunities for study and recreation of which you have never before thought of availing yourself; and the treasures of South Kensington and the National Gallery, which you have hitherto neglected as 'slow' and 'bad form,' are now a source of delight to you. The only fault that you can now find with the latter institution is, that it spoils you for all the modern galleries about Pall Mall and Piccadilly. You have a feeling of proprietorship now in the royal parks, which you never had when you sauntered in the Row, or attended the meet of the Coaching Club at the Magazine, or dawdled about the Mall in St James's Park on a Drawing-room day. You don't attend these 'functions' now, for, though they are open to you as to the rest of the world, you feel yourself rather out of the race. But you often enjoy the air in the higher ground of Hyde Park, which you will come to consider as bracing as the Sussex Downs; nor are you to be persuaded that Burnham Beeches has a much finer show of trees than Kensington Gardens.

But the time when you do really and thoroughly enjoy the Pleasures of Ruin is when that delectable moment comes—which it inevitably will, sooner or later—when a temporary, or, let us hope, it may be a permanent, change in your fortunes takes place. Your book has found a publisher; your picture a buyer; some one pays up an old debt; or an unknown relative mentions your name in his will. Whatever it may be, the keen appreciation of the benefits we formerly enjoyed which our vicissitudes have taught us, and the knowledge we have acquired of the dingier side of nature, give a remarkable zest to our return to a brighter life. And if a man has good health and good spirits, he will find that it is as true that 'hope springs eternal in

the human breast,' as that when things are at their worst they mend; and if he is of an extra-hopeful disposition, he will welcome the increased depression of his fortunes as a sure forerunner of a change of luck.

COUSIN GEORGE.

IN TWO CHAPTERS.—CHAP. II.

ALL went well in the Smethby circle, indeed things had never before gone so smoothly in that not unprosperous group. Harriet, it is true, did not get more manageable in the Robert Crewe direction; she was perfectly ready to flatter and please the Australian cousin, and had an eye to the main chance as keen as others; but the young doctor was not to be jeopardised. Thus Harriet might be regarded as an exception; so, of course, might Mr Crewe; but after all, as he does not actually appear in our narrative, he need not count for much.

There were frequent indications that the ridiculous disguise, the absurd plea of poverty, at first put forth by Styles was being gradually discarded—was ‘peeling off,’ Mr Joe said, with a happy touch of description. But Mr Smethby would not see all these indications—pretended not to notice any flaws; he would humour his cousin just as long as the latter chose.

The proposed investment was still in favour, was about to be made, indeed; and so earnest was Cousin George in the matter, that when Smethby said he had given notice at the bank for his money, he confidentially told him that if there was any difficulty about getting it, his friend would advance the sum for a week or two—or for a year, if Smethby would like it. The latter thanked him, but declined. Of course he could see through this, as he had seen through the other flimsy screens.

The bank was good enough, he explained, and so it was, for the money was duly paid to him; and it was proposed that they should go up to town together, Smethby and Cousin George, where the latter would see his friend’s broker and arrange for the purchase of this stock.

In a confiding mood, not usual with him, Smethby had proposed that Styles should send a cheque up, or go up with it by himself, if going up were necessary; but the latter declined to do this. He seemed to have a strange dislike to cheques or drafts, and as he said: ‘It was not their way at the

diggings; a man liked to look after his own business there.’ So Cousin Nick must go with him.

He, Cousin George, had also asked Harriet what kind of bracelet she preferred; for his friend had desired him to consult some lady’s taste, as he, the friend, was thinking of making a little present. Harriet was not proof against this temptation, so explained that amethyst bracelets with amethyst pendants—or sapphire and diamonds, if she *did* have her choice—was what she liked. Cousin George, with a highly expressive wink on hearing this, said his friend would be much obliged by her opinion. He should perhaps see him on the next day but one when he, Styles, and her father went to London.

‘All which means, my dear,’ said Smethby, when he had a chance of whispering to his daughter, ‘that this farce is about to end. He means to present me with the whole of these twenty thousand shares, and you will have a present also. Beyond this, you will have an offer in plain language—his language has already been plain enough to show what he means; so, be a sensible girl, and don’t lose a chance the like of which will not occur again, if you live for a hundred years.’

Harriet did not reply; there was indeed a recurrence of the pouting and flouncing; she could not resist the jewelry; but when Robert Crewe was endangered, she exhibited some of the old perverseness.

In the morning, Cousin George took a stroll into the town, as was his habit. Smethby knew quite well that his eccentric relative went to the post-office, whither his letters, as every one knew, were directed. No one, however, pretended to suspect anything like this arrangement, which was just as shallow and easily penetrated as his other schemes. On his return, he was in higher spirits than usual; a little fitful, perhaps, but certainly more jocular and fuller of sly allusions than he had hitherto allowed himself to be. This was evidence enough, to such a man as Smethby, to show that the end of the scheme was approaching. He broached a capital joke—he undoubtedly so considered it—in the way of a question as to what his cousin Nick would have thought of and said to him, Styles, if he had come back from the diggings loaded with shiners—‘Not one or two, Nick, but some scores of thousands, eh!—what then, Nick?’ he exclaimed.

Smethby was of course acute enough to seize such a palpable chance, so replied with the utmost heartiness and frankness, that, delighted as he should have been at such good fortune, it never could have made any difference in his feelings to his old friend and cousin, George Styles. The latter grasped his hand at this, and seemed for the moment almost overcome by his feelings. He was indeed about to say something, which Smethby expected would prove a clearing-up avowal; but he checked himself, and saying abruptly, 'No; wait a day or two,' turned the conversation.

Yet, all through the day, there was an uneasiness in Cousin George's manner which could not escape the attention of those around him; and he took several short strolls in the open air to soothe his nerves, which, he admitted, seemed rather shaky. On the last occasion that he took his saunter, it was in the twilight, and in the glance which he naturally threw around him before entering the house, he could see, standing in relief against the clear summer sky, the figures of two men, who were apparently conversing earnestly as they paused on a knoll not far from Mr Smethby's residence.

Then Styles went in, and found the lamps were just lighted, the curtains were drawn, while his host and his daughter, evidently in the best of moods, were awaiting him. With a decision which was almost like abruptness, Styles began about the visit to London on the morrow. He explained, as he had done before, that until the transaction was completed, he did not want any one, not even the broker, to know that the stock was not entirely for his friend, who had promised to take over all the disposable shares; and that was why he had asked Mr Smethby to provide money instead of a cheque for the payment.

'I understand,' smiled Smethby; 'and, as you know, I have arranged to get notes in the morning. But here is the cheque, if that would suit you—you can have it to-night, if you like.'

'No; O no!' returned Styles; but the response came so slowly, that it seemed as if he had hesitated before deciding. 'There will be no use in that; so long as I can see the broker alone, that will do.'

'Just as you please,' said Mr Smethby. As he paused, a ring at the street door was heard.

‘And now a word or two about that little villa my friend thought of buying at Richmond,’ resumed Styles. ‘I had a letter this morning’——

‘If you please, sir,’ said the maid-servant, appearing at the door, ‘a gentleman wishes to see you.’

‘To see me, or to see Mr Styles?’ asked her master. Another ring was heard at the street door as he said this.

‘I believe I want to see both of you,’ said a voice behind the servant, which voice being deep and harsh in its tone, and coming so unexpectedly, made each person in the room start; ‘so I shall take the liberty of coming in here,’ continued ‘the gentleman;’ then, suiting the action to the word, he pushed past the attendant, and came close to the table which filled the centre of the room.

All looked at him in amazement; while, before any one spoke, Mr Joe and Mr Brooks, who had called just then to have a chat with Mr Styles, also entered, and gazed at the stranger with as much astonishment as was shown by their friends. The stranger was an elderly, grizzled, but powerfully built man, with hard features, high cheek-bones, indented nose, square jaws, hidden by his stiff iron-gray beard, and moustache.

‘You are Mr Smethby—Nicholas Smethby, I believe: in fact, I know it,’ said the man.—‘But may I ask who this is?’ pointing to Cousin George as he spoke.

‘I really do not know what your business here is, or why you make this inquiry,’ returned Smethby, a good deal nettled by the intrusion; ‘but I certainly am Nicholas Smethby, and this gentleman is Mr George Styles. Have you any business with either of us?’

‘Did you ever see George Styles look like a cross between a skittle-sharp and a stage smuggler?’ continued the visitor, ‘which is what this fellow looks like.’

‘Do you mean’—— began Cousin George, but he spoke falteringly; while Mr Joe and Mr Brooks, who stood behind the stranger, could see that the speaker turned pale.

‘Yes; I do mean,’ interrupted the visitor; ‘and I mean a good deal more than that, as you will find.’ He flourished an ugly-looking stick which he carried,

as if to give emphasis to these words.—‘As for you, Nick Smethby, I am surprised and ashamed to think you could be such a fool as to mistake a fellow like this for your own cousin—for *me!*’

Here every hearer started in reality; and Smethby, drawing a long breath, looked from one to the other with an expression which clearly showed that he did not mean to contest the announcement.

‘Do you think,’ resumed the new-comer, ‘that a man, after twenty years’ beating about the diggings, which I have had, could look as young as he did when he started? which is pretty nearly what this fellow does, in spite of his make-up.—I have come back with enough to pay you your loan, Nick, but I have been down very low in my time. I have fought two battles in the colonial ring, and I am going to show this fellow, presently, how I won them.’

‘All this is dreadfully mysterious!’ exclaimed Smethby; ‘yet one thing is clear enough: I will swear you are my cousin George Styles. But then, who is this?—Yes, who are you, you impostor?’ he cried, turning sharply upon his guest, who gasped once or twice, as though trying to speak, but was paralysed by the new-comer, from whom he could not remove his eyes.

‘Don’t trouble yourself about him yet,’ pursued the second Styles. ‘I will just say what I have to say, and then I will get it all out of him; you will see that. I fancy, however, I am only just in time. Is it true that you have agreed to go up to London with this person and invest a lot of money among his confederates?’

The ‘first cousin,’ as he may fairly be called, groaned at this; while Mr Smethby uttered, as well he might, an ejaculation of intense astonishment at finding his intentions and plans thus known to a man whom he had not seen for twenty years.

‘I see you are surprised, Nick, and that our customer there feels he is bowled out,’ said the stranger. ‘But after all, there is nothing to wonder at in the matter. I inquired my way at the station—having learnt your address from your old office—and a gentleman who overheard me, kindly offered to show me the place. I told him who I was; and he was just as much as flabbergasted as you are; but he was delighted as well. He told me all about this’—— The speaker paused while he cast a look of utter contempt at his

predecessor, and then went on, evidently unable to find an epithet suitably strong. 'He told me he was a doctor, by name Robert Crewe.' (It was now Harriet's turn to start and change colour.) 'We walked together to a point just below here, where he turned off at the brow of a hill. He not only told me about the impostor who was taking my name, but pointed him out as he slunk in at the gate.' (The unlucky cousin remembered, and groaned audibly as he did so, the two men whom he had seen in converse on the rise in the road.) 'So here I am; and the first thing I mean to do is to collar this fellow, and thrash him until he has not a sound inch of skin on his carcase.—But don't you turn pale, my dear.' This was said to Harriet, and the speaker raised his cap with a sort of reassuring politeness. 'Though I have come straight from the mines, I do not forget what is due to a lady; and I shall take the fellow outside to have his thrashing, and he shall have it now.' With this, he made a stride forward, and thrusting his huge hand inside the man's collar, clutched him with a grip which might have been of iron, and with a single tug pulled him to his feet; but the victim seemed unable to stand, and sank back on his chair all of a heap.

Harriet uttered a scream as the real Cousin George bent over the man, evidently intent upon dragging him out by main force; while Mr Joe and Mr Brooks seized his arm, and urged him not to be violent—Joe at the same moment briefly introducing himself and his brother-in-law.

'I am glad to see you again, anyhow, young Joe,' returned Styles. 'I remember buying you a drum the last time I was in your company.—But you had better let me settle this fellow at once.'

'Spare me!' whined the man. He could not speak comfortably with such a grip on his collar and with such knuckles buried in his neck.

'Why, what I am going to do is real mercy to you!' retorted his captor. 'You will be sore for a week or ten days, and then be as well as ever; but if I give you over to the police—— Well, as you seem to dread a simple licking so much, we will go to the police. Come on!'

Another tremendous tug here dragged up the unfortunate creature, who broke into most despairing petitions, imploring that they would not give him up to the police—*they* knew him, he said.

‘Why, confound it! you do not suppose you are to be let off scot-free, after such a game as this, do you?’ exclaimed the other, whose astonishment was so clearly genuine, that Joe and Brooks could not repress a smile.

‘I will confess everything; I throw myself on your mercy!’ urged the man; ‘but don’t give me up to the police. I am sure to get it hot, if you do.’

‘So you ought!’ ejaculated Styles.

‘I think if you were to quit your hold on his neck, he could speak freer,’ said Mr Joe; ‘and I should really like to know how all this came about.’

‘Ah! so he might,’ assented Styles, acting on the suggestion. ‘I can easily catch hold of him again when I want him. I’ll bet he does not give us the slip.’

In spite of the threat conveyed in the last speech, the culprit’s face visibly brightened after Joe’s remark. Mr Smethby had remained silent all this time, being not only confused with the unexpected revelation, but a little ashamed, possibly, of his own management, which was so over-cunning as to make him a readier prey to the swindler.

‘Well, go on,’ was the rough command of Styles. ‘Who are you? Where do you come from?’

‘My name is John Smith,’ began the man. A furtive leer which he cast upon the company as he said this, might have been involuntary; but certain it is that none of those who saw it believed he was speaking the truth. ‘I had got into trouble,’ he continued, ‘and wanted some money for a fresh start. While I was at my wits’ end to get this, a pal—a friend—who knew I had been in a difficulty, said’ (he paused here, and glanced at Smethby)—‘he said there was a flat to be had at Valeborough, if he was properly worked.—No offence, I hope, sir. It was not me who said this; it was my friend.’

‘It was correct enough, whoever said it,’ replied Smethby, to whom the remark had been addressed.

‘He knew a lot about the family affairs here,’ continued Smith: ‘he had scraped about and picked the particulars up, till he thought he had got quite enough to enable a man to act as the cousin they had not seen for twenty years; but he owned he had not got the headpiece to keep the game up for any time; so I was to be the cousin; and he was to be a friend who knew me,

and was to manage—as he did very well—to get hold of Mr Smethby, as if by accident, and tell him all about the good luck of his old friend Styles, and how he was going to try on a game with his cousin Mr Smethby.’

‘I never thought I was such an idiot; but go on,’ said the host.

‘We raked up some money between us,’ resumed Smith; ‘but it was a hard job to get enough, as of course I had to be pretty liberal; but luckily this gentleman would not let me spend much.—However, I got a letter this morning, saying that Ben—my friend—could not send another penny, and that unless I could make a haul at once, the thing must burst up. But the business was nearly ripe. I had prepared the way for persuading my cousin, as I called him, to invest a lot of money, by dropping a pretended letter from my stockbroker, which I knew they would find and read. In fact, there was no difficulty all through; and I had arranged for a visit to London tomorrow, so I was in hope that’——

‘That you could make the haul,’ said Smethby, as the other paused. ‘How did you mean to do it, when I should be with you? I was to go to the office, you know.’

‘I meant to take you to a place where you would wait in a room, while I went into what you would think was only an inner office, but which I knew had a way out,’ answered Smith. ‘In fact, if I had once touched the money, there would have been an end of it.’

‘And your friend with the villa and the bracelets?’ asked Smethby.

‘All put in to make it seem more natural,’ said the man. ‘But I have not robbed your place of a pennyworth ever since I have been here, I assure you. I hope you will take that into consideration.’

He went on a little further, until he was interrupted by Styles, who led him to the door—no force was now wanted—and telling him that he would give him in charge to the nearest policeman if he ever saw him again, pitched him out on the dark road, and then returned to the circle he had left.

At first, Smethby was terribly chopfallen, but recovered ere long, and joined in the laugh with which first ‘Cousin George’ and then the others reviewed the past. Harriet was not the noisiest of the party, but she was not the least happy, and ‘Cousin George’ appeared to have taken a great fancy for her.

Styles paid his debt to 'Nick Smethby' that night, to prove, as he said, that he was not another impostor, and said, besides, that while he should not bother about amethyst bracelets or diamonds and sapphires, yet, if that young doctor had the courage to get married within three months, and a few hundreds would help him to get into practice, why, he George Styles, had enough for such a purpose, and Harriet should take care of it, until it was wanted.

Altogether, although rougher and coarser than the first cousin, this second edition was a great improvement; and settling down as he did in Valeborough, he was a regular visitor, not only at Mr Smethby's but at Dr Crewe's, when the latter set up his own house, after an early marriage to Miss Harriet.

And improvident and wild as George had once been, he was steady enough in his friendships now, so he never left the little circle; and when he died, his property—a good deal less than the hundreds of thousands attributed to the first cousin—went to the children of Dr and Mrs Crewe, with which cluster of young people he had always been a great favourite.

AIR AS A MOTIVE FORCE.

IN a recent number of the *Journal* we touched on the various methods of transmission of power, and showed how steam had been laid on in mains in the streets of American towns, and a house-to-house distribution thus effected. Loss has been found, however, to result from leakage and condensation, and these defects have militated against the system. Water under pressure has obtained extended application in this country where power was required in docks and warehouses; but up to the present time, a motor has not been introduced satisfying the necessary requirements of economy sufficiently to render the system of commercial value for supplying small power either for domestic purposes or to the lesser industries. Bursting of pipes, through frost or other cause, might result in serious damage, moreover, in dwelling-houses.

The problem of transmission of power may possibly find a solution in electricity in the future; but as regards the present, suffice it to say that the cost of production of such agency entirely precludes it from entering into the field of competition. Attempts now being made, in Paris and Birmingham, to distribute power by rarefied air in the former, and by compressed air in the latter city, possess no slight interest. In each case, the method adopted differs in no way in principle from that of the systems already touched on. Central pumping stations, furnished with boiler and steam-power, supply the requisite energy; whilst the transmitting medium—steam, water, or air, as the case may be—is distributed through the principal mains, which feed in their turn the lesser arteries of the system supplying the individual consumer.

In the case of rarefied air, though, theoretically, a pressure of fifteen pounds per square inch could be obtained, in practice it is found advisable to work at a pressure of about ten pounds, without approaching nearer to an absolute vacuum. Three classes of motors are employed to convert the vacuum in the mains into useful work; suffice it to say, however, that whilst differing in the details of construction, the principle involved throughout is the same, and consists essentially of modifications of the steam-engine to the

requirements of air-pressure. Payment is made according to the power absorbed by each consumer, an ingenious arrangement actuating as counter, indicating how much work is actually done, irrespective of the number of revolutions made by the motor. Even where gas is available, the cost of engines for using it has not unfrequently militated against its adoption by the smaller industries; hence the Parisian Company for the distribution of power by rarefied air has elected not only to supply power but to lease out the motors as well. Their customers embrace such users of small power as hat-block makers, jewellers, wood-turners, comb-cutters, stay and clothing manufacturers, dentists, butchers, &c. The cleanliness of this system, and its excellent ventilating capabilities, should form an argument in its favour. Not only is all smell from combustion, as in the case of the gas-engine, avoided, but, by drawing at every stroke a given quantity of air from the room, the motor directly produces ventilation.

Time alone can show whether the system will prove a commercial success; in any case, its promoters could hardly have chosen a better field for its introduction than Paris, a city containing upwards of a million persons engaged in the minor industries already indicated, and which require small motive power.

A NINETEENTH-CENTURY PIRATE.

IT is not likely that many of our readers will have heard of a certain Captain Hayes, who a few years ago was one of the most notorious desperadoes among the numerous 'beachcombers' and other questionable characters who infested the South Pacific. A few instances of this worthy's escapades in the paths of fraud and villainy, drawn from *Coral Lands*, by H. S. Cooper (London: R. Bentley & Son), may be of interest, and will also show how, up to a comparatively recent period, a determined character could pursue a career of actual crime and piracy in the Eastern seas with impunity.

Of the antecedents of Captain (or 'Bully,' as he was commonly dubbed) Hayes, little is known before 1858, when he appeared in the Hawaiian Islands, having landed from the ship *Orestes*. After a short stay at Honolulu, he left for San Francisco in the beginning of 1859; and a few months afterwards reappeared in command of a brig bound for New Caledonia. Having entered a closed port without having first passed the custom-house, the sheriff arrested him and took possession of the brig. Captain Hayes put all the blame on his first officer, and was virtuously indignant with him for misinforming him as to the necessity of first entering at the custom-house at Lahaina, at the same time treating the sheriff with unbounded courtesy and every mark of respect. He at once agreed to proceed to Lahaina, and seemed delighted to find it was the sheriff's duty to accompany him thither. When, however, the ship was clear of the land, Hayes 'changed his tune,' and coolly informed the sheriff he had no intention of going near the custom-house, and that he (the sheriff) could either remain on board and pay for his passage to New Caledonia, or find his way back to port the best way he could. The sheriff found himself completely outwitted, and was perforce obliged to take to his small boat—luckily, still alongside—and managed to reach the land with considerable difficulty, having the melancholy satisfaction of seeing his late prisoner laughing at him over the taffrail as he resumed his course for the Southern Ocean. Next mail brought instructions to the United States consul at Honolulu for Hayes' arrest; and it then became known that when last in the islands he had borrowed money from a confiding clergyman, with which he had gone to San Francisco and

negotiated the purchase of the brig, fitted her out, engaged his crew and then set sail, paying nobody. His cruise at this time, however, did not last very long; shortly afterwards, his ship was wrecked at Wallace's Island, the captain and his 'chums' escaping in the boat to the Navigators' Islands, leaving the rest of the crew to their fate. They ultimately, however, succeeded in getting safe to shore by means of a raft.

Hayes was next heard of at Batavia in command of a barque; how obtained is not known. He succeeded in getting a cargo of coffee for Europe—which it would never have seen—when the Dutch East India Company got some information as to his antecedents, and were only too glad to get repossession of their coffee, losing the charter-money, which Hayes insisted on being paid before he allowed the cargo to be taken on shore again. Finding he had not much chance of doing any good—or evil, rather—at Batavia, Hayes resolved to depart in search of a fresh field for the exercise of his talents. Proceeding to Hong-kong, he succeeded in filling his vessel with Chinese coolies, and sailed for Melbourne. After a fair voyage, he was nearing the Australian coast, when he spoke a ship, and was informed that a tax had been imposed on all Chinese immigrants, and that he would have to pay fifty dollars per head on his passengers before he would be permitted to land them. This was rather a serious outlook for the captain, but, as usual, his inventive brain was equal to the occasion. He sailed calmly on, and soon arrived off his port of destination. Then he set to work to carry out the plan he had conceived. He coolly filled his ship half-full of water, hoisted signals of distress, and lay to, waiting the development of his ruse. He had not long to wait; his signals for assistance were perceived, and two tug steamers were soon alongside, proffering their services for the purpose of towing him into port. Hayes declared his ship would sink before she could be got into dock, as his pumps were choked and the water rising at a great rate. He implored them to take off his passengers, leaving his crew and himself to escape by means of their boats, should the barque not float till they returned. This the tug-owners agreed to do. The Chinamen were transhipped, and the steamers bore off, promising to return as speedily as possible to his assistance. They got their load of Chinamen safely landed, the owners paying the head-tax, and steamed back to bring in the ship; but she was nowhere to be seen, having, as they supposed, gone down with all hands. No such fate, however, had befallen the gallant captain. No sooner were the tugs out of sight, than he pumped his ship free of water, and lost

no time in putting a good few miles between him and Melbourne, inwardly chuckling, no doubt, at the clever way he had duped the antipodeans and got his Chinamen landed at others' expense. Some time after this, Hayes speculated in another cargo of Chinamen; but this time he landed them without trouble and without paying anything, having gone through the formality of getting them all made British subjects before he sailed!

For a few years after this, Captain Hayes was little heard of, except at some of the South Pacific islands, where he occasionally turned up, ostensibly pursuing the avocation of an honest trader. By-and-by, however, he resumed his old habits, and for a couple of years or so he made raids on several of the island groups, robbing and destroying the stations of the traders and native villages. Eventually, he was arrested by the British consul at Upolu. As luck would have it, at this same time a certain friend of Hayes, Captain Pease or Peace, arrived at Upolu in his brig the *Leonora*. On some pretence or other, Hayes obtained leave to go on board; and when next morning dawned, the brig was invisible, having sailed during the night with him on board as a passenger. In due time, the *Leonora* arrived at Shanghai, and by some dodge or other, Hayes managed to get Captain Pease put in prison, passing himself off to the authorities as the owner of the brig. He next got on board the supplies he was in need of, and set sail, as usual paying for little or nothing. Hayes once more was in command of a good ship, with a crew who asked no questions, and in a position to resume his fraudulent career. His first port of call was Saigon, where he was chartered to take a load of rice to Hong-kong and other intermediate ports. At the first port of call, the owner of the rice went on shore to try and effect a sale. Hayes took this opportunity of leaving the owner behind, and set off for Bankok, where he disposed of his cargo at a good price, and departed once more for his favourite hunting-ground—the South Pacific.

Hayes some time after this was again without a ship, having imprudently intrusted his vessel to the care of his first officer, who treated the 'Bully' to a dose of his own game, and went off with her, leaving him in a quandary on one of the South Pacific islets. Hayes was now forced to change his play, and accordingly came out in a new character. Pretending to be converted from his evil ways, he completely got the better of the American missionaries, and obtained command of a small schooner belonging to the Mission. At the first favourable opportunity, as may be supposed, he

disappeared with the schooner, and arrived at Manila. Here, however, his fame had preceded him, and on being recognised, he was promptly arrested, and put in prison. The captain's game seemed now about up; but his good luck had not yet deserted him. Once more adopting the religious dodge, he turned a devout Catholic, and so talked over the priests, that, although there was evidence enough to hang him and a dozen others besides, he got off, and was next heard of at the scene of his first escapade, San Francisco, where he stole a smart schooner called the *Lotus*, and once more was off for the Sunny South.

On another occasion, Hayes was captured by the U.S. steamer *Narraganset*, which had been commissioned to look out for him. He was not many days on board the war-ship, when, by his affable manners and gentlemanly behaviour, he so won over the sympathies of the American officers, that they became convinced he was a most worthy individual, and set him free, actually supplying him with a new set of sails and other articles he was in need of!

On another occasion, Hayes called at Levuka, the capital of Fiji, to obtain supplies for a lengthened cruise. The goods were sent on board, and the bill rendered, payment being expected next morning before he sailed; but when the day dawned, the captain, as usual, was off. Unfortunately for him, however, in this instance the wind failed him, and the merchant was able to overtake the ship in a rowboat.

The captain was not at all put about when the merchant came on board; said 'he presumed he would have letters for him to post, and would be delighted to be of use.' The merchant was rather taken aback at such coolness in an absconding debtor, and mildly hinted at payment of his account.

'Why,' exclaimed Hayes, 'you were paid yesterday!'

The merchant assured him that he was mistaken.

Hayes expressed astonishment, and ordered up one of his officers. 'Didn't I give you the cash to settle this gentleman's bill?' he asked indignantly; and then the 'Bully' opened the vials of his wrath upon the innocent seaman, who was cunning enough to see the captain's object, and held his tongue. Seeing, however, that there was no sign of a breeze springing up, he was

forced to pay for his supplies, no doubt very much chagrined at having to be honest for once in his lifetime.

After a long career of robbery and bloodshed—for he gets the name of having perpetrated several murders—Hayes at last met his deserts at the hands of one of his officers, whom he had defrauded and ill-used in a most disgraceful manner. No doubt, the secret of his eluding the hands of justice for so long a time was his particularly pleasing manners and appearance. He was by no means a common ruffian, but the reverse, having a handsome face and figure, and bestowing a deal of care and attention on his personal appearance. His urbanity of manner and conversational powers were of the most fascinating description, and he could entertain a friend or knock him on the head in an equally charming style. When he first appeared in the Pacific, he was accompanied by ‘Mrs Hayes,’ and was seldom without a female companion, several of whom are said to have been among his victims. He was possessed of great natural abilities. If he had only turned his talents into a proper channel, he might have made a good position for himself in the world.

THE MONTH: SCIENCE AND ARTS.

MR C. TANKERVILLE-CHAMBERLAIN, late acting consul at Panama, gives a hopeful account of the progress of M. de Lesseps' giant undertaking, the construction of the Canal across the Isthmus, which is very different from the description of the state of things lately published in the American newspapers. He believes that the great work will be actually completed in about three years' time. The line of the Canal, forty-six miles in length, has been divided into five sections, which have been handed over to five responsible and solvent contractors, who are bound under heavy penalties to complete their work by the end of 1888. The holders of railway stock and many others in America are interested in believing, and trying to make others believe, that the Canal is a failure and cannot succeed. That it will be a financial success, must remain an open question, for the expense already incurred, added to that which is to come, constitutes a larger sum than has ever yet been sunk in a single engineering undertaking.

A proposal is now on foot to connect by means of a submarine tunnel the defences of Portsmouth with the forts on the Solent and with the Isle of Wight, and it is probable that preliminary borings will be made to ascertain the practicability of the scheme. It has been before proposed that a fort should be built half-way between Stokes Bay and Ryde, on a bank which rises to within eight feet of high-water mark; but the scheme was abandoned because of the difficulty of finding fresh water for the garrison. The tying together of this proposed fort and the other defences would at once obviate this difficulty, and would at the same time relieve our expensive ironclads from the duty of protecting a spot which has always been looked upon as of great importance.

Among all the wonderful things which were exhibited in the late Colonial and Indian Exhibition, there was nothing more remarkable than the vast variety of different woods—strange to European eyes—which were shown in some of the Courts. These woods seemed to exhibit every shade of colour and every variety of grain. In one Court in particular could this be well

remarked, for the different samples of wood were cut into the shape of books and highly polished, each pseudo volume bearing its own name. Messrs A. Ransome & Co. lately invited a number of colonial visitors—engineers, builders, and others—to their large works at Chelsea, in order that they might demonstrate the applicability of some of these woods to various purposes. About forty different varieties were subjected to the operations of tree-felling, cross-cutting, sawing, planing, moulding, mortising, tenoning, and boring; while various articles, from casks to doors, were actually made and completed before the visitors' eyes. The exhibition not only formed an illustration of the suitability of many colonial woods for employment in this country, but it also showed to what a marvellous pitch of perfection wood-working machinery has been brought by Messrs Ransome. The demonstration is likely to lead to a great shipment of colonial woods to this country, many of which are plentiful, and therefore cheap.

The colossal statue of Liberty, which has been presented by the French Republic to the Republic of America, and which, with the pedestal, is over one hundred and fifty feet in height, is, at the time we write, nearly completed. When the statue is quite finished, it is proposed to illuminate it at night in a very novel manner. The female figure of Liberty holds aloft a torch, which will be furnished with eight electric arc lamps, each of six thousand candle-power, the rays from which will be thrown upwards towards the clouds. At the same time, several other lamps of similar power will shine on the statue itself, causing it to stand out in strong relief from its dark surroundings.

A correspondent of the *Times*, quoting a letter recently received from Sydney, New South Wales, gives an account of the extraordinary instinct shown by ants and other insects which live in and on the ground. Some months ago, the natives of a certain district predicted the approach of floods, and left their low-lying camping-grounds for the higher country. The floods came as predicted, several weeks later; and the natives said that their sole information regarding them was gathered from the insects, which had built their nests, &c. in the trees, instead of, as usual, in the ground. The correspondent asks whether this forecasting providence of the ant is recorded by any of our travellers, and whether any explanation of the fact can be given.

Here are two more natural-history notes recorded by correspondents. It is pointed out by one that, owing to our backward spring this year, the swallows on their arrival were kept so short of food that quite two-thirds of their number died of famine; hence the unusual plague of flies that we have experienced during the summer. He pleads that the little mud nests which are seen clinging under the eaves of so many houses in country and suburbs should be protected from injury, for if it were not for the swallows, flies would constitute a veritable pest.

In answer to this, another writer points out that sparrows will sometimes prevent the swallows building, and will often drive the rightful owners from their nests. This fact he has ascertained by direct observation. He also remarks that the swarms of flies this year may be due in great measure to the scarcity of wasps, which destroy an immense number. The scarcity of wasps in his particular neighbourhood is fully accounted for, one of his friends having destroyed no fewer than sixty-seven of their nests. His plan of procedure is, as far as we know, as novel as it is simple and effective. Tow soaked in spirits of turpentine is thrust into the wasp's nest at night, and the hole is afterwards filled up—presumably with earth.

We are so accustomed to wonderful news from the land of Niagara, that we are not much surprised to learn that the largest photographic negative ever produced has been taken by an American worker. The glass plate upon which the colossal picture was taken measured sixty by thirty-six inches, and weighed more than eighty pounds. The coating with sensitive material of such a plate was in itself a very difficult undertaking, while for its development after exposure in the camera, over three pailfuls of fluid had to be cast over its surface while it was lying in a specially constructed tray. The photographer succeeded in obtaining a good picture, as well as a silver medal to reward him for his enterprise.

A French journal says that flowers may be preserved with all their natural brilliancy and freshness by dipping them into a mixture made as follows: In a well-corked bottle, dissolve six drachms of coarsely powdered clear gum-copal; add the same quantity of broken glass, and fifteen and a half ounces (by weight) of pure rectified sulphuric ether. The flowers should be dipped into this varnish-like fluid four or five times, allowing them to remain in a current of air for ten minutes between each immersion. This plan, if it does not interfere with the delicate texture of the petals, should be of use to

flower-painters, who often have to hurry their work unduly because of the perishable nature of their models.

Mr Graber has lately made some curious observations upon the effect of light upon eyeless animals, a Report of which appears in the Proceedings of the Vienna Academy. He put a number of earthworms into a box, which was provided with an aperture at one side, through which light was allowed ingress. The result of many experiments showed that the worms sought the darkest part of their temporary prison, and that at least two-fifths of their number shunned the light. Experimenting with rays of different colours by means of stained glass, he found that the worms exhibited a marked preference for red light.

According to the *American Druggist*, an alloy which will solder glass, porcelain, and metals, or one to the other, can be made in the following manner: Copper dust, made by precipitating the metal from a solution of bluestone by means of zinc, is put into a mortar and treated with strong sulphuric acid. To this mass, formed by the copper and acid, is added a little more than twice as much mercury, the addition being made with constant stirring. The amalgam thus formed is washed with warm water to remove the acid, and is afterwards cooled. When required for use, it is heated, and worked in a mortar until it becomes as soft as wax, and in this state it will cling tenaciously to any surface to which it may be applied. It is applicable more especially to those substances which will not bear a high temperature.

A year ago, Mr J. W. Swan of Newcastle described before the North of England Institute of Mining and Mechanical Engineers an electric safety-lamp which he had invented for the use of miners. This lamp, although efficient, had no means of detecting the presence of firedamp. In an improved lamp which the same inventor has produced, this deficiency is supplied, for a firedamp indicator forms part of the lamp. This indicator is based upon one invented some time ago, and consists of a coil of platinum wire which can be switched on to the current which supplies the lamp and brought to a red-heat. If firedamp be present, the wire becomes far hotter, and therefore brighter than it will in pure air; and in one form of lamp a similar coil, shut up in a glass tube containing air, is provided, for the sake of comparison. In another form of indicator the hot wire is made to explode the charge of firedamp submitted to it, of course in a closed chamber, thus forming a partial vacuum, which acts upon a column of liquid in an

attached gauge tube. By this means the exact percentage of fiery gas present can be accurately noted. It may be hoped that these improved appliances may come into common use; but of course electrical fittings are somewhat expensive, and this is the initial difficulty in introducing improvements which would lead to much saving of life.

In these enlightened times, when books without number are published to instruct even the youngest scholars about the nature of common things, it seems almost impossible to realise the ignorance which existed and the nonsense which was written even as lately as the last century concerning matters of the most elementary kind. So-called facts in natural history of the most ludicrous kind were handed down from writer to writer and accepted as the exact truth by all readers. Here is a specimen of chemical knowledge which dates from the year 1747, and is due to the pen of one George Adams. He naively remarks that 'some people have imagined that the sharpness of vinegar is occasioned by the eels striking their pointed tails against the tongue and palate; but it is very certain that the sourest vinegar has none of those eels, and that its pungency is entirely owing to the pointed figure of its salts, which float therein.' There is probably some confusion here between the sourness of vinegar and the acidity of sour paste, which latter is accompanied, as even young microscopists know well, by the development of innumerable so-called eels.

At a recent meeting of the Society of Medical Officers of Health, Dr Alfred Hill, the President, delivered an opening address, which dealt with the important subjects of the disposal of house-refuse and the best method of treating sewage. The employment of destructive furnaces for getting rid of dry house-refuse was strongly recommended. The efficient disposal of sewage is of course a far more difficult problem to solve, and one which has now for a number of years troubled the minds of many. Dr Hill is in favour of the sewage-farm principle, which has been so successfully tried at Birmingham. He showed that the system had not proved a nuisance to adjoining residents nor yet injurious to health. It was also a profitable system, for in the city referred to, twenty thousand pounds had been realised during the past year by the sale of stock and produce from the sewage-farm. He believed that if a similar system were adopted for the metropolitan area, the sewage which is now allowed to poison the Thames might realise in meat, milk, and vegetables two hundred thousand pounds.

Mr Thomson Hankey has lately pointed out a new use for sugar, which, however, is not new, but it is so little known that he has done good service in calling attention to it. In the preparation of mortar and cement, the addition of a certain quantity of unrefined sugar will give the mixture extraordinary hardness and tenacity. In India, sugar has been used for this purpose from time immemorial, and walls built with mortar of this description will defy all ordinary methods of destruction. Plaster of Paris will also set much harder if about ten per cent. of sugar be added to the water with which it is mixed. With plaster of Paris, it might be mentioned, the addition of alum has much the same effect.

At one of the recent meetings of the Iron and Steel Institute, M. Gautier of Paris read an interesting paper on 'The Casting of Chains in Solid Steel.' In the course of this paper, he pointed out that in order to compete successfully with wrought-iron in chain-making, the steel employed must be quite solid and absolutely free from blowholes, and it is most necessary to adopt a quick method of moulding the chains. In the process which has been adopted by Messrs Joubert and Leger of Lyons, these difficulties have been successfully overcome. The process combines chilled casting with instantaneous removal from the moulds, after which the chain is finished and annealed in oil. By this method he claims that better chains can be manufactured than those of wrought-iron, with the advantage of greatly diminished weight.

The deposition of dust and smoke by the passage of electricity has been more than once adverted to in these pages, more especially in connection with the collection of lead-fume. Messrs King, Mendham, & Co. of Bristol have recently constructed a convenient piece of apparatus for illustrating this phenomenon. It consists of a jar capped at the top with a cover, through which protrudes a rod furnished with a ball. This rod terminates inside the jar in a point; and a similar pointed wire, which finds a termination outside the lower part of the jar, is opposite to it. Below, there is a small combustion box, in which a smouldering piece of brown paper will soon fill the jar with smoke. Thus filled, the jar is connected by its brass terminals to a Wimshurst Electrical Machine. When the handle of the machine is turned, an electrical discharge takes place between the two pointed wires; and the smoke, after being violently agitated, disappears, leaving the air in the jar perfectly clear.

The Simplex Ironing Machine, which is invented by Mr S. Bash, and which has been examined and approved by the leading tailoring establishments in London and Paris, is designed to relieve workers from the heavy manual labour attending the use of pressing-irons. The simplex iron is suspended from a movable arm by a universal joint, and can be moved in any direction over the work and with any desired degree of pressure. This pressure is brought about by the aid of a pedal attachment. There is also provision made for pressing long seams, a movable table being made to travel to and fro beneath the gas-heated iron. The inventor claims for his method a saving in fuel and more rapid and efficient work.

A new explosive has been invented by a Russian engineer, M. Rucktchell, about which some very curious particulars have been published, while the nature of the compound remains the secret of its discoverer. The explosive gives a penetrative power to projectiles ten times greater than gunpowder. It emits neither smoke nor heat, and its discharge is unaccompanied by any report. If this be true, can the compound—whatever it be—be called an explosive? But this wonderful product is to be utilised in the arts of peace as well as those of war, for it forms the motive-power for an engine constructed by the inventor, an engine for which he claims superiority over steam and gas engines. It will be remembered that an engine of much the same character was invented a few years ago in America. Its motive-power was a secret from everybody. The necessary and inevitable Company was formed to buy up the inventor's rights, and then—nothing more was heard of it.

Mr W. F. Dennis has been exhibiting at Millwall, London, a continuous wire-netting machine, which is a great improvement on former contrivances of this kind. The machine works from bobbins of wire only, not from bobbins and spools, as in the older machines, and these bobbins contain a sufficient length of wire to keep the machine at work for a whole day. In a day of ten hours, a single machine will produce three hundred and fifty yards of wire-netting twenty-three inches in width. The machine in question occupies a space of eleven by eight feet, by six feet in height. Nor is it confined to the production of netting from soft metal, for hard bright steel and iron wire can be used, producing a most rigid product. The consumption in Europe of wire-netting is estimated at forty million yards

per annum, and the possibility of producing it of a rigid character, hitherto thought to be impossible, is sure to increase its fields of usefulness.

OCCASIONAL NOTES.

WOODITE.

WOODITE, a newly invented preparation of caoutchouc—so called from the name of its inventor—is attracting considerable attention at the present time. In woodite are united the useful elastic properties of india-rubber together with the advantages of immunity from injury by fire or salt water. The specific gravity of woodite is only one-tenth that of iron or steel; whilst the cost of the new material, as compared with these metals, is estimated to be as three to seven, or rather less than one half. Such facts fully explain the importance attached to the proposition now being made to utilise woodite as a protection—either internal or external, as regards the vessel's skin—to men-of-war and torpedo boats. Experiments recently made to ascertain the behaviour of woodite under fire were as satisfactory as conclusive, and established the interesting fact, that the caoutchouc closed up again so thoroughly and instantaneously, after the passage of the shot, that no leakage resulted, though the vessel was pierced below waterline.

The value of a material possessed of such qualities for naval purposes cannot be overestimated; whilst in a variety of other ways, woodite appears likely to play a not unimportant part in the near future. In the construction of lifeboats, a material so buoyant and indestructible cannot fail to be of service; whilst for lining quay walls, harbour entrances, piers, landing-stages, and the numberless cases where it is desirable to moderate the force of impact, woodite should be found of the greatest value. In the case of a collision at sea, a vessel fortified internally or externally with woodite would be more likely to remain afloat, than, *cæteris paribus*, one not similarly protected.

In an age when every effort is made to secure the requisite buoyancy in our huge floating citadels, heavily laden with ponderous armour and gigantic ordnance, a material combining buoyancy in so high a degree, with its other advantages, cannot but be destined, in the opinion of competent judges, to

play a brilliant part; whilst its future in the more peaceful arts cannot fail to be equally commensurate with its merits.

TRAVELLING ARRANGEMENTS ON THE CANADIAN PACIFIC RAILWAY.

A passenger by the Canadian Pacific Railway gives an interesting sketch of the travelling arrangements on this latest trans-continental line. We learn that the locomotives have a haul of about one hundred and twenty to one hundred and thirty miles in each division of the line, when they are changed, and fresh ones put on. The continent is crossed from Montreal to Vancouver, in British Columbia, in five days and fourteen hours; and this will soon be reduced to one hundred and twenty hours. Good time is kept. The first east-bound trans-continental train that was met in transit, passed Sudbury, going eastward, at 4.17 P.M., after being about five days on the journey. Before its arrival, there was some curiosity to learn whether it was in time, and bets were made on the time it would arrive. This train, after travelling a distance of two thousand five hundred miles, arrived only fifteen seconds behind time. The railway route from Montreal to Vancouver covers two thousand nine hundred and nine miles; and the through sleeping-coaches attached to the train run the entire distance without change, which is a great comfort to the traveller. Every week-day, a train starts from each end of the line, leaving the eastern terminus at Montreal at eight o'clock in the evening, and the western terminus at one o'clock in the afternoon. On Sundays, the trains do not start; thus making six trains each way every week. The west-bound train is called the Pacific Express; and the east-bound train the Atlantic Express.

The Pacific Express, in which this correspondent travelled, was made up of five coaches. At the head was the luggage, mail, and express coach, which carried the baggage. The next is the colonists' coach, a third-class carriage with seats arranged so that they can be turned into a double tier of berths on each side for sleeping accommodation. The train carries passengers at three rates. The ordinary American first-class passenger coach follows the colonists' coach, which usually takes local travellers along the line. Following this is the dining-coach, which usually accompanies the train only from seven o'clock in the morning till nine o'clock at night. Following

the dining-car is the through sleeping-coach, which is constructed with six sections on each side. In the aggregate, twenty-six persons can be given sleeping accommodation in this car; while at one end, toilet-rooms and a bathroom are provided. At the rear of the sleeping-coach is a large open apartment with a good outlook, which can be used as a smoking-room, and where passengers may have a view of the line passed over.

OVERHEAD TELEGRAPH WIRES.

This arrangement of wires has always been considered as a disfiguring and dangerous eyesore, and at last our quick-sighted cousins 'across the water' have determined that the nuisance shall be forthwith abated. In New York, Washington, St Louis, Chicago, and other great cities of the United States, legislative decrees have been issued for the compulsory abolition of all overhead wires, which will in future be conducted underground in tunnels beneath the pavement, and by this means a great improvement will be effected in the matter of street architecture, and some dangers to passengers will be removed. Many instances have been known in America where, from violent storms of wind or snow, the telegraph posts have been blown down, occasioning injury and even death to passengers. All this will be avoided by the new arrangement.

ANGRY BEES.

As a supplementary note to the article on '[Bees and Honey](#)' which appeared in [No. 135](#) of the *Journal*, a correspondent sends us the following:

'A painful instance of the terrible consequences of provoking bees is connected with one of the loveliest sights in India, the famous Marble Rocks of Jubbulpore. These rocks form a gorge through which the great river Nerbudda flows, and the marble formation extends for about a mile. The dazzling walls which shut in the river are studded with pendent bees' nests, and for any one proceeding in a boat down the narrow channel to disturb the bees is a fatal proceeding. If any warning were required, it is given by a tomb which stands on the outskirts of the village just above the gorge, to the memory of one who was

stung to death in this beautiful spot. Actuated by a foolish impulse, he fired his rifle at one of the nests, whereupon the bees came down on him in such numbers that he attempted to save himself by jumping overboard. The relentless insects, however, still pursued him, with fatal results. I quote the story from memory, but believe it is to be found in detail in Forsyth's charming work, *The Highlands of Central India*.

‘A friend once told me that as he was driving near a village some miles from Jubbulpore, he and his servant and horse were attacked by bees without any real provocation. The enemy crowded round in such numbers that the situation became serious. After receiving several stings, and finding the horse, too, becoming restive, my friend resolved to save his own life and that of his servant, both of which were really in jeopardy, at the risk of a little discomfort to other people. Accordingly, he whipped up his horse and made for the village, a cloud of bees keeping up with the trap without the least effort. When the village was reached, the bees, as my friend anticipated, found so many other objects of interest, that they distributed their attentions with less marked partiality than hitherto. In other words, the cloud left the trap and scattered among the villagers, who were, however, so numerous, that two or three stings apiece probably represented the total damage. The expedient was not, perhaps, a charitable one, but, in the circumstances, was, I venture to think, justifiable.’

The PUBLISHERS have pleasure in intimating that next year will appear in this JOURNAL an Original Novel, entitled

RICHARD CABLE,

by the distinguished Author of the well-known works of fiction, ‘Mehalah,’ ‘John Herring,’ ‘Court Royal,’ &c.

A BRIGHT DAY IN NOVEMBER.

A SUMMER hush is on the golden woods;
The path lies deep in leaves—the air is balm;
No sound disturbs these silent solitudes,
Save some faint bird-notes, which, amid the calm,
Seem like the sad, sweet song of one who grieves
Over a happy past—yet with a strain
Of Hope, which sees amid these yellow leaves,
Bare boughs all clothed with Spring's young buds again.

Even thus, most gracious Lord, in Sorrow's hour,
When Life seems saddest, and our hopes decay,
Thou sendest comfort—as, in wood or bower,
Some humble flower remains to speak of May;
Some gleam of joy lights up the wintry scene;
Some tender grace returns to bless and cheer;
And though our trees no more are clothed in green,
Bright days may light the closing of our year.

J. H.

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[Transcriber's note—the following changes have been made to this text.

Page 764: Naraganset to Narraganset.]

*** END OF THE PROJECT GUTENBERG EBOOK CHAMBERS'S
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